Modeling and Analysis of Information Leakage under Multiple Adversaries

Lichen Fu and Wenye Wang
Department of Electrical & Computer Engineering
NC State University
Raleigh, NC 27606, USA
Emails: {lfu3,wwang}@ncsu.edu

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***Abstract*—The rapid growth of IoT devices and applications has resulted in the generation and transmission of large volumes of private data, which encompass various dimensions and are observable by multiple entities. To address the need for balancing confidentiality and data utility, the Quantitative Information Flow (QIF) approach has been developed to quantify information leakage in computer systems. However, it assumes a *single* adversary seeking to infer a *single* piece of secret information. This paper examines leakage measures in the Multiple Attributes and Adversaries (MAA) scenario where attributes are under coordinated attacks from multiple adversaries through individual side channels. The paper reveals that the current model, when applied individually to each attribute or adversary, only considers scenarios where all attributes are leaked to all adversaries. However, any leakage of an attribute to any adversary should be deemed a breach of private data. To tackle this, the paper introduces the MAA leakage model and validates its compliance with a leakage measure and operational interpretation of the MAA threat model. Additionally, it presents a closed-form expression for the upper bound of MAA leakage within a given set of side channels, demonstrating that the upper bound is achieved when all attributes conform to a uniform distribution.**

***Index Terms*—Data leakage, attributes, multiple adversaries.**

I. Introduction

Protecting the confidentiality of sensitive information while preserving its utility is a fundamental security concern. To address the disclosure of information that relies on sensitive data, research focuses on quantifying acceptable levels of information leakage. The field of Quantitative Information Flow (QIF) theory in computer systems emerges to regulate the transmission of sensitive information from one discrete random variable (denoted as X) to another (denoted as Y). Here, the secret is represented as X with a prior probability distribution π, while the adversary's observation is represented as Y with a posterior distribution. Initially, Shannon entropy [1] was extensively employed to quantify information leakage, as it measures uncertainty. The entropy $H(X)$ represents the information within X, and the reduction of entropy $H(X) - H(X|Y)$ gauges the information leakage. Later, Smith [2] introduced the concept of *vulnerability* as the worst-case probability of correctly guessing the value of X in a single attempt, measured using min-entropy.

Drawing upon Information Theory for theoretical analysis, the existing QIF framework necessitates treating secrets as

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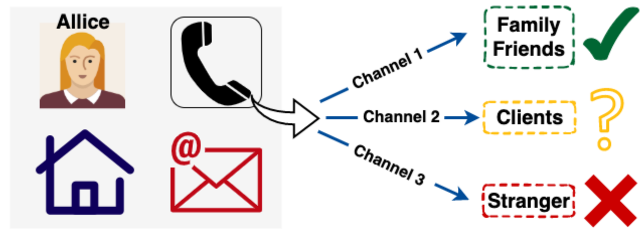

Fig. 1. A User Record with Multiple Attributes.

random variables. This, in turn, gives rise to streamlined assumptions regarding the structure of the data. However, real-world private data often exhibit complex structures, such as medical records, online shopping habits, or GPS location. Frequently, these data attributes depict discrete yet interconnected traits that collectively paint a comprehensive portrayal of a data entity. As an example, prominent data breaches [3] exposed various attributes of user profiles, potentially exposing data owners to cyberattacks. By modeling secrets as random variables, the existing QIF framework is limited to examining one attribute at a time [2], [4], [5], thereby lacking the capacity to comprehensively encapsulate the significance and interconnections of leakage across these attributes. Consequently, a pressing need arises to develop models that encompass data with multiple attributes and seamlessly integrate them into established techniques for privacy leakage analysis.

The QIF framework originates within the domain of computer systems, where research often centers on a solitary system (channel) and can efficiently extend to more intricate systems through the amalgamation of single systems. However, quantifying data leakage grows complex when confronted with scenarios encompassing multiple adversaries operating through distinct channels. We consider data leakage as the unauthorized disclosure of sensitive information, where the concept of "unauthorized" varies based on specific data attributes. For instance, personal details like date of birth and home address may be sensitive to the general public but not to family members, while trade secrets may be discussed among business teams but kept confidential from friends. Additionally, data leakage occurs whenever data is transferred, and any recipient of the data can be considered a potential adversary for certain data attributes. This highlights the inevitability of data leakage in the context of necessary data sharing and exchange, underscoring the need to regard all data recipients as potential adversaries with different leakage severity. On the other hand, it is crucial to recognize that a system can have multiple exploitable side channels, these factors must be considered during the design of cryptographic algorithms and protocols to mitigate risks posed by multiple adversaries.

In this paper, we propose a modeling approach to address the aforementioned scenarios by incorporating attribute leakage and considering the physical implications of a coordinated multi-adversary attack model. Our contributions are:

- We are the first to consider privacy leakage in the MAA scenario. We identify the limitations of the current QIF model and analyze the prerequisites of an accurate leakage model in such scenarios by examining its physical interpretations. This analysis establishes a sturdy groundwork for affirming the new model's inherent naturalness.
- We introduce MAA leakage, a generalization of min-entropy leakage, and validate its correctness as a leakage measure in the MAA threat model.
- We derive the capacity of MAA leakage for a given set of channels, providing a closed-form expression that is validated through numerical results.

The remaining sections of this paper are organized as follows: Section II provides a brief introduction to QIF. Section III discusses the necessary requirements of the new model under the MAA threat model. Section IV presents the definition of the new model and its properties. Section V analyzes the capacity of MAA leakage and presents simulation results. Finally, Section VI concludes the paper.

II. Preliminary

In the field of computer security, it is often necessary to tolerate a certain level of confidential information leakage. To address this, theories within the QIF have been developed to determine the acceptability of certain leaks. Early research in QIF draws upon information theory [1] in communication systems, where information leakage is quantified as the reduction in uncertainty of the secret, measured by $H(X) - H(X|Y)$, which corresponds to the Mutual Information $I(X;Y)$. In this context, we provide a brief overview of information-theoretic channels, vulnerability, and min-entropy leakage.

A. Information-theoretic Channel

A system can be mathematically represented as a communication channel characterized by a tuple $(\mathcal{X}, \mathcal{Y}, C)$. Here, $\mathcal{X}$ and $\mathcal{Y}$ denote finite sets representing the possible inputs and outputs, respectively. The channel matrix C is a $|\mathcal{X}| \times |\mathcal{Y}|$ matrix, where each element $C_{x,y}$ represents the probability of the system producing output y when the input is x.

Consider two random variables, $X : \Omega \to \mathcal{X}$ and $Y : \Omega \to \mathcal{Y}$, defined on a shared probability space Ω. If the probability distribution P_X and the channel matrix C are known, we can compute the joint distribution P_{XY}, marginal probabilities P_Y, as well as conditional probabilities $P_{Y|X}$ and $P_{X|Y}$.

B. Vulnerability and Min-entropy

According to Smith [2], a system with lower mutual information is more susceptible to being guessed by an adversary in a single attempt compared to a system with higher mutual information. Shannon entropy, based on averaging, fails to capture the risk associated with this single event with high probability. Consequently, it is not suitable for providing a security guarantee. To address this limitation, Smith introduces the concept of *vulnerability* for a random variable X and defines leakage based on min-entropy.

Definition 1 ([2])**.** *Given a random variable X, the vulnerability of X, denoted $V(X)$, is given by*

$$V(X) = \max_x p(x). \tag{1}$$

Given jointly distributed random variables X and Y, the conditional vulnerability $V(X|Y)$ is

$$\begin{aligned} V(X|Y) &= \sum_y p(y) V(X|Y=y), \text{ and} \\ V(X|Y=y) &= \max_x p(x|Y=y). \end{aligned} \tag{2}$$

The min-entropy $H_\infty(X)$ and conditional min-entropy $H_\infty(X|Y)$ are defined as

$$\begin{aligned} H_\infty(X) &= -\log V(X), \text{ and} \\ H_\infty(X|Y) &= -\log V(X|Y). \end{aligned} \tag{3}$$

The concept of vulnerability captures the probability of correctly guessing the secret in a single attempt, and it is quantified using the measure of min-entropy in bits. The information leakage of secret variable X is determined by the reduction in min-entropy when observing Y. The notions of additive and multiplicative approaches to information leakage, along with their capacities, have been studied in [4], [5].

III. Problem Formulation

Existing QIF approaches overlook the complexity of real-world scenarios where secrets involve multiple attributes and observations of multiple adversaries. We present two examples to illustrate the lack of applicability of QIF metrics and the desired properties of a leakage metric tailored for such cases.

A. What are data attributes?

Data attributes are distinct characteristics used to describe or identify a data entity. Protecting the confidentiality of private data involves securing all its attributes and considering attribute-specific leakage when assessing overall data leakage.

In this paper, we adopt the assumption that attributes are independent and represent them as a set of random variables denoted as $\mathbf{A} = \{X_1, X_2, ..., X_K\}$, accompanied by a corresponding distribution $\boldsymbol{\pi} = \{\pi_1, \pi_2, ..., \pi_K\}$.

Example 1. *Consider a user record Δ consisting of four attributes $\Delta(\mathbf{A}) = X_N, X_P, X_H, X_E$, representing Name, Phone Number, Home address, and Email shown in Fig. 1. The record Δ is observed by one adversary and denoted as Y. How do we describe the leakage of Δ?*

To align with the QIF model's focus on single secrets, we can treat each attribute as an individual secret and analyze their information leakage independently. This assumes attribute independence, allowing us to obtain the leakage of Δ by summing the leakages of its attributes.

$$\mathcal{L}_{min}(\mathbf{A} \to Y) = \log \frac{\prod_i V(X_i|Y)}{\prod_i V(X_i)}, \tag{4}$$

where $i \in \{N, P, H, E\}$. The numerator represents the probability of an adversary correctly guessing all attributes in a single attempt, given the observation Y, while the denominator represents the probability of an adversary correctly guessing all attributes through a blind guess.

However, this definition is flawed as it assumes an ”all or nothing” scenario, where leakage occurs only when the adversary gains complete access to all attributes. In reality, any form of leakage, whether involving some attributes or partial information, should be considered as leakage of the record Δ. Therefore, the numerator and denominator cannot accurately represent the posterior and prior vulnerability of Δ, which we denote as $\mathcal{V}(\mathbf{A}|Y)$ and $\mathcal{V}(\mathbf{A})$, respectively.

Additionally, it's important to note that $\prod_i V(X_i|Y) \leq V(X_i|Y)$ and $\prod_i V(X_i) \leq V(X_i)$, as the vulnerability is measured as a probability ranging from $(0,1]$. This inequality may seem counter-intuitive when considering the leakage of the record. Since the leakage of any attribute contributes to the leakage of the record, the confidentiality of a record is more vulnerable than that of an individual attribute. Therefore, $\mathcal{V}(\mathbf{A})$ and $\mathcal{V}(\mathbf{A}|Y)$ should follow the following relationship:

$$\mathcal{V}(\mathbf{A}) \geq V(X_i), \text{ and } \mathcal{V}(\mathbf{A}|Y) \geq V(X_i|Y). \tag{5}$$

B. What are the channels?

Before delving into the multiple adversaries model, it is essential to establish the framework for the channels through which attributes are observed by different adversaries.

1) A Uniform Channel: One approach to modeling the data channels is to consider them as a single uniform channel. In this case, multiple adversaries receive their observations through the same information-theoretic channel denoted as $P_{Y|X}$, where the observation is represented by the random variable Y. This operational scenario aligns with previous works of QIF, such as those discussed in [1], [2], [6].

Example 2. *If the secret $X \sim Uniform$ is observed by three adversaries as y_1, y_2, and y_3 through a Binary Symmetric Channel (BSC) with parameter p ($0 \leq p \leq \frac{1}{2}$). Here, y_1, y_2, and y_3 can take any arbitrary values from the alphabet of Y.*

- Shannon entropy: $1 + (1-p)\log(1-p) + p\log p$.
- Min-entropy leakage: $\log 2(1-p)$.
- Maximal leakage: $\log 2(1-p)$.

2) Individual Channels: In Fig. 2, it is evident that Eve and Mallory may not share the same channel due to potential changes and disturbances introduced through their connection. Thus, it is more appropriate to consider them as separate channels within the context of multiple adversaries.

We propose modeling each adversary's observation as being received through an individual channel. For a secret X, the i-th adversary's channel is represented as $C_i = P_{Y_i|X}$, where Y_i is the random variable representing the observation made by the i-th adversary. We assume that each adversary's observation has a finite and fixed alphabet, regardless of the input.

Consider the case where N adversaries observe K attributes through N channels. We define a matrix of channels denoted as $\mathbf{C} = \{\mathbf{C}_{i,j} = P_{X_i|Y_j} \mid 1 \leq i \leq K, 1 \leq j \leq N\}$. Here, $\mathbf{C}$ is a collection of $K \times N$ channel matrices, where each matrix describes the relationship between a specific attribute and an adversary's observation. In the following subsection, we will demonstrate that conventional QIF methods concentrate

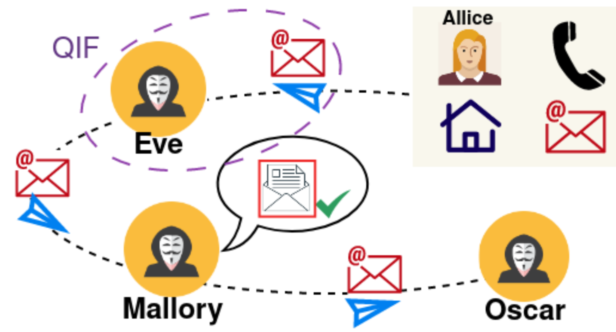

Fig. 2. A User Record and Multiple Adversaries.

solely on a single channel, as illustrated inside the circled dashed area, and are ill-suited for the intricate system depicted in Fig. 2. In contrast, our approach adopts a simplified and manageable modeling strategy tailored to MAA scenarios.

C. What are multiple adversaries?

An adversary is someone who aims to gain unauthorized access to confidential information, which can include external hackers as well as insiders with legitimate data access. We assume multiple adversaries may collaborate after receiving their observations through independent side channels. We model N observations as a set denoted as $\mathbf{O} = \{Y_1, Y_2, ..., Y_N\}$.

Example 3. *Given the prior vulnerability $V(X_E)$ of the attribute X_E and the observations $\mathbf{O} = Y_E, Y_M, Y_O$ from three adversaries (Eve, Mallory, and Oscar) as depicted in Fig. 2, the adversaries update their posterior vulnerability as $V(X_E|Y_E)$, $V(X_E|Y_M)$, and $V(X_E|Y_O)$. How can we characterize the leakage of X_E?*

To model multiple adversaries intuitively, each adversary can be treated independently, applying QIF to their individual observations. This assumes the independence of the adversaries. Consequently, the total leakage of X_E can be obtained by summing the leakages observed by each individual adversary.

$$\mathcal{L}_{min}(X_E \rightarrow \mathbf{O}) = \log \frac{\prod_i V(X_E|Y_i)}{\prod_i V(X_E)}. \tag{6}$$

where $i \in \{E, M, O\}$. The sum of individual adversary leakages may overestimate the actual leakage of X_E. Min-entropy leakage quantifies leakage based on the logarithm of the increase in probability. For example, if Y_E and Y_M each increase the vulnerability of X_E by 0.1 and 0.2 respectively, the total increase in the probability of correctly guessing X_E is 0.2, not 0.3. Hence, the sum of individual leakages does not accurately represent the actual leakage of attribute X_E.

Additionally, it's worth noting that the sum of leakage across all adversaries can be arbitrarily large. However, the leakage of X_E under multiple observations $\mathbf{O}$, denoted as $\mathcal{L}(X_E \rightarrow \mathbf{O})$, is upper bounded by its min-entropy.

$$\mathcal{L}(X_E \rightarrow \mathbf{O}) \leq H_{min}(X_E). \tag{7}$$

The numerator in Eq. (6) represents the probability of all adversaries correctly guessing the secret in a single attempt after their observations, while the denominator represents the probability of all adversaries correctly guessing the secret in a blind guess. This definition is flawed as it assumes that leakage occurs only when all adversaries collectively guess the correct secret value. However, any leakage to any adversary should be considered as a leakage of the secret. Therefore, the numerator and denominator cannot represent the posterior and prior vulnerabilities of X_E, denoted as $\mathcal{V}(X_E|\mathbf{O})$ and $\mathcal{V}(X_E)$.

Moreover, it is true that $\prod_i V(X_E|Y_i) \leq V(X_E|Y_i)$ and $\prod_i V(X_E) \leq V(X_E)$, indicating that leaking to all three adversaries is less likely than leaking to any single adversary. However, when observed by multiple adversaries, the vulnerability of X_E is larger compared to being observed by a single adversary, as leakage to any adversary contributes to the overall leakage of X_E. This approach underestimates $\mathcal{V}(X_E|\mathbf{O})$. The vulnerabilities $\mathcal{V}(X_E)$ and $\mathcal{V}(X_E|\mathbf{O})$ should satisfy the following relationship:

$$\mathcal{V}(X_E) = V(X_E), \text{ and } \mathcal{V}(X_E|\mathbf{O}) \geq V(X_E|Y_i). \tag{8}$$

Lastly, it is important to acknowledge that the existing QIF framework has limitations in modeling the collaboration among multiple adversaries. The framework assumes independence of observations and does not consider the potential impact of joint attacks or coordinated efforts among adversaries.

D. What are requirements of MAA leakage?

If $\Delta =< \mathbf{A}, \mathbf{O} >$ represents a piece of data with K attributes $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$ and is observed by N adversaries $\mathbf{O} = \{Y_i \mid 1 \leq i \leq N\}$ through individual channels, we propose that an appropriate leakage metric should satisfy the following requirements:

(R1) It should provide an intuitive physical interpretation:

$$\mathcal{V}(\mathbf{A}) \geq V(X_i), \tag{9a}$$

$$\mathcal{V}(\mathbf{A}|\mathbf{O}) \geq \mathcal{V}(X_i|\mathbf{O}), \tag{9b}$$

$$\mathcal{V}(\mathbf{A}|\mathbf{O}) \geq \mathcal{V}(\mathbf{A}|Y_i), \text{ and} \tag{9c}$$

$$\mathcal{L}(\mathbf{A} \to \mathbf{O}) \leq H_{min}(\mathbf{A}). \tag{9d}$$

(R2) Assumptions about the adversaries should be kept minimal. The fewer assumptions we make about the adversary, the more broadly applicable the leakage model becomes.

(R3) It should align with intuition, meaning it should accurately capture the information leakage in systems that we understand well, without mischaracterization.

IV. A Multiple Attributes and Adversaries (MAA) Model with Side Channels

In this section, we introduce the MAA model that captures the relationship between data and attributes concerning leakage. This model aligns with an intuitive understanding of the vulnerability in a collaborative multi-adversary attack scenario.

A. System Model

We define a piece of data as a tuple $\Delta :=< \mathbf{A}, \mathbf{O} >$, where $\mathbf{A}$ represents the set of attributes and $\mathbf{O}$ represents the set of observations. Each attribute in $\mathbf{A}$ is modeled as a separate random variable $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$. For each attribute i, we have a probability triple $(\Omega_i, \mathcal{F}_i, P)$, where Ω_i is the finite sample space, $\mathcal{F}_i$ is the associated σ-algebra, and P is the probability measure defined on $\mathcal{F}_i$.

To illustrate, let's consider the attribute "gender" with the probability triple $\Omega = \{Male, Female\}$, $\mathcal{F} = \{\varnothing, \{Male\}, \{Female\}, \Omega\}$, and the probability measure $P(\{\omega \in \{Male\}\}) = P(\{\omega \in \{Female\}\}) = \frac{1}{2}$.

Given a measurable space $(E_i, \mathcal{E}_i)$, where E_i is a set and $\mathcal{E}_i$ is the corresponding σ-algebra, a random variable X_i is a function $X_i : \Omega_i \to E_i$ that satisfies $X_i^{-1}(B) \in \mathcal{F}_i$ for all $B \in \mathcal{E}_i$. The probability measure of X_i, denoted as π_i in the case of $E_i = \mathbb{R}$, is defined as $P(X_i \in B) = P(X_i^{-1}(B))$, where $B \in \mathcal{E}_i$. This probability measure is commonly referred to as the "prior" in the context of QIF [7]–[9].

The observations are represented as a set $\mathbf{O} = \{Y_i \mid 1 \leq i \leq N\}$, where Y_i corresponds to the observation made by the i-th adversary. For each adversary's observation, there is a probability triple $(\Phi_i, \mathcal{C}_i, P)$, and the random variable $Y_i : (\Phi_i, \mathcal{C}_i) \to (D_i, \mathcal{D}_i)$. The conditional distribution $P(Y_j|X_i)$ describes the relationship between the observation made by the j-th adversary and the i-th attribute.

B. Modeling with Multiple Attributes

We now develop a theory of data leakage that incorporates the attributes of the data. The objective is to model the attributes as an integral part of the data, ensuring that any leakage of an attribute contributes to the overall data leakage.

Definition 2. *Given a set of random variables of K attributes $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$, the prior vulnerability is given by*

$$\mathcal{V}(\mathbf{A}) = \sum_{i=1}^{K} \max_{x_i \in \mathcal{X}_i} P(X_i = x_i). \tag{10}$$

Each term in the summation represents the maximum probability that an adversary can correctly guess the value of an attribute within one attempt. The aim is for $\mathcal{V}(\mathbf{A})$ to capture the overall vulnerability of all the attributes, surpassing the vulnerability of any individual attribute. However, it should be noted that $V(\mathbf{A})$ is not a probability.

An alternative approach that may be suggested is to consider $\mathcal{V}(\mathbf{A}) = \max_i V(X_i)$ and $\mathcal{V}(\mathbf{A}|Y) = \max_i V(X_i|Y)$, which selects the worst-case scenario among the attributes to represent the leakage of Δ. However, this approach only focuses on a single attribute and does not account for the leakage of other attributes. To illustrate, consider the following example:

Example 4. *A user's record has two attributes, that are, $V(X_1) = 0.2$, $V(X_2) = 0.6$, $V(X_1|Y) = 0.6$, and $V(X_2|Y) = 0.6$. If we take the maximum value among all attributes as the posterior and prior vulnerability, the leakage of the record is 0, while there is leakage at the first attribute.*

Following the same concept, we define the posterior vulnerability of the data as the sum of the worst-case probabilities across attributes. This ensures that the posterior vulnerability reflects the collective vulnerability of all attributes, while still surpassing the posterior vulnerability of each attribute.

Definition 3. *Given joint distributed random variables $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$ and Y, the posterior vulnerability is*

$$\mathcal{V}(\mathbf{A}|Y) = \sum_{i=1}^{K} \sum_{y \in \mathcal{Y}} P(Y = y) V(X_i|Y = y), \tag{11}$$

$$\text{where} \qquad V(X_i|Y = y) = \max_{x_i \in \mathcal{X}_i} P(x_i|Y = y). \tag{12}$$

Definition 4. *Given data $\Delta =< \mathbf{A}, Y >$ with K attributes observed by one adversary as Y, the leakage of Δ is given by*

$$\mathcal{L}(\mathbf{A} \to Y) = \log \frac{\mathcal{V}(\mathbf{A}|Y)}{\mathcal{V}(\mathbf{A})}. \tag{13}$$

This definition offers a quantitative measure of the increased likelihood of accurately guessing any attribute within the data following an adversary's observation. The definition ensures that if $\mathcal{L}(\mathbf{A} \rightarrow Y) \leq l$, the combined probability of accurately guessing each attribute is limited by $2^l \sum_{i=1}^{K} \max_{x_i} P(x_i)$. The system designer can not control the term $\max_{x_i} P(x_i)$, but is able to determine l through the channel matrix $P_{Y|X}$.

Remark 1. *The definition of data leakage under multiple attributes inherently aligns with (R1) Eq. (9a) and Eq. (9b).*

In Example 1, with the joint distributions $P_{X_N Y} = (0.4, 0.2; 0.3, 0.1)$, $P_{X_P Y} = (0.3, 0.2; 0.1, 0.4)$, $P_{X_H Y} = (0.4, 0.3; 0.2, 0.1)$, and $P_{X_E Y} = (0.4, 0.4; 0.1, 0.1)$, where $V(X_P|Y)$ increases by 0.2 compared to $V(X_P)$ and the other attributes exhibit no leakage, we observe that $\mathcal{L}_{min}(\mathbf{A} \rightarrow Y)$ is approximately 0.485, while $\mathcal{L}(\mathbf{A} \rightarrow Y)$ is approximately 0.107. The $\mathcal{L}_{min}$ is larger as it represents the leakage when the adversary successfully guesses all attributes simultaneously, primarily driven by the leakage of X_P in this case. On the other hand, $\mathcal{L}$ takes into account the increased vulnerability of each attribute as part of the leakage measure, resulting in a smaller value that reflects the leakage of the record Δ.

C. Modeling with Multiple Attributes and Adversaries

In this section, we examine the data leakage model when multiple adversaries target various attributes. Following (R2), we assume that each adversary receives observations through independent channels and can collaborate and share information before attempting to guess a set of attributes. To begin, we introduce the posterior vulnerability under MAA.

Definition 5. *Given joint distributed random variables* $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$ *and* $\mathbf{O} = \{Y_i \mid 1 \leq i \leq N\}$, *the posterior vulnerability* $\mathcal{V}(\mathbf{A}|\mathbf{O})$ *is*

$$\mathcal{V}(\mathbf{A}|\mathbf{O}) = \sum_{i=1}^{K} \sum_{\mathbf{o} \in \mathbf{O}} P(\mathbf{O} = \mathbf{o}) \mathcal{V}(X_i | \mathbf{O} = \mathbf{o}), \tag{14}$$

$$\text{where} \quad \mathcal{V}(X_i|\mathbf{O} = \mathbf{o}) = \max_{x_i \in \mathcal{X}_i} P(x_i | \mathbf{O} = \mathbf{o}). \tag{15}$$

$\mathcal{V}(X_i|\mathbf{O})$ represents the probability of successfully guessing attribute X_i based on observations $\mathbf{O}$. Conditional probability $p(x_i|\mathbf{o})$ captures the adversary collaboration. While the side channels ($P_{Y_i|X}$) are independent, the variables Y_i are dependent on each other through input variable X. However, for $i \neq j$, conditional independence holds: $P(Y_i|Y_j, X) = P(Y_i|X)$.

The definition of MAA leakage is given as follows:

Definition 6. *Given* $\Delta =< \mathbf{A}, \mathbf{O} >$ *with* K *attributes* $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$ *observed by* N *adversaries* $\mathbf{O} : \{\mathbf{O}_{i,j} = Y_i^j \mid 1 \leq i \leq K, 1 \leq j \leq N\}$, *the leakage of* Δ *is given by*

$$\mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) = \log \frac{\mathcal{V}(\mathbf{A}|\mathbf{O})}{\mathcal{V}(\mathbf{A})}. \tag{16}$$

This definition quantifies the increase in the probability of correctly guessing any attribute when observed by multiple adversaries. It incorporates the advantage gained by any adversary in guessing the value of any arbitrary attribute into the leakage measurement of Δ. The definition ensures that if $\mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) \leq l$, the cumulative probability of correctly guessing each attribute is limited by $2^l \sum_{i=1}^{K} \max_{x_i} P(x_i)$.

In Example 3, multiple adversaries observe a single attribute data with joint distributions $P_{X_E Y_E} = (0.4, 0.2; 0.1, 0.3)$, $P_{X_E Y_M} = (0.4, 0.2; 0.3, 0.1)$, and $P_{X_E Y_O} = (0.1, 0.5; 0.2, 0.2)$. X_E shows a vulnerability increase of 0.1 for both Y_E and Y_O. We find that $\mathcal{L}_{min}(X_E \rightarrow \mathbf{O}) \approx 0.445$ and $\mathcal{L}(X_E \rightarrow \mathbf{O}) \approx 0.245$. $\mathcal{L}_{min}$ overestimates the leakage by summing vulnerability increases at each observation, while $\mathcal{V}_{X_E|\mathbf{O}} \approx 0.711$ exceeds the maximum posterior vulnerability from a single observation, demonstrating additional information gained through collaboration.

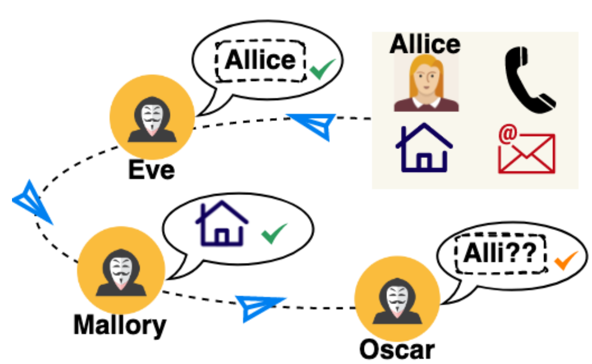

Fig. 3. Multiple Attributes and Adversaries.

Example 5. *Consider the same example with a record* $\Delta =< \mathbf{A}, \mathbf{O} >$ *as depicted in Fig. 3. The record is shared among three adversaries: Eve has full access to* X_N, *Mallory has full access to* X_A, *and Oscar possesses partial information about* X_N. *Assuming* $V(X_N) = 0.3$, $V(X_P) = 0.2$, $V(X_H) = 0.5$, *and* $V(X_E) = 0.3$, *we can determine that* $\mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) = \log[(1 + 0.2 + 1 + 0.3)/(0.3 + 0.2 + 0.5 + 0.3)] \approx 0.943$.

We prove MAA leakage aligns with (R1) Eq. (9c) and (9d).

Lemma 1. *For any* $\boldsymbol{\pi}$ *and* $\mathbf{C} : \{\mathbf{C}_{i,j} = P_{X_i|Y_j} \mid 1 \leq i \leq K, 1 \leq j \leq N\}$, $\mathcal{V}(\mathbf{A}|\mathbf{O}) \geq \mathcal{V}(\mathbf{A}|Y_i) \geq \mathcal{V}(\mathbf{A})$, $1 \leq i \leq N$, *which implies that MAA leakage is non-negative.*

Proof. The observations are independent given the secret X_i:

$$\begin{aligned} \mathcal{V}(\mathbf{A}|\mathbf{O}) &= \sum_{i=1}^{K} \sum_{\mathbf{o}} \max_{x_i} p(x_i, \mathbf{o}) \\ &\geq \sum_{i=1}^{K} \sum_{y_j} \max_{x_i} p(x_i, y_j) = V(\mathbf{A}|Y_j), \ \forall 1 \leq j \leq N \\ &\geq \sum_{i=1}^{K} \max_{x_i} \sum_{y_j} p(x_i, y_j) = \sum_{i=1}^{K} \max_{x_i} p(x_i) = \mathcal{V}(\mathbf{A}). \ \square \end{aligned}$$

Lemma 2. *MAA leakage is upper bounded by the average mean of all the attributes' min-entropy.*

Proof. From Jason's inequality, we can show that:

$$\begin{aligned} \mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) &\leq \log \sum_{i=1}^{K} \sum_{\mathbf{o}} \sum_{x_i} p(x_i, \mathbf{o}) - \log \sum_{i=1}^{K} \max_{x_i} p(x_i) \\ &= \log K - \log \sum_{i=1}^{K} \max_{x_i} p(x_i), \end{aligned}$$

$$\log \frac{\sum_{i=1}^{K} \max_{x_i} p(x_i)}{K} \geq \frac{\sum_{i=1}^{K} \log \max_{x_i} p(x_i)}{K},$$

$$\mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) \leq \frac{1}{K} \sum_{i=1}^{K} H_\infty(X_i). \quad \square$$

Theorem 1. *Given* $\Delta =< \mathbf{A}, \mathbf{O} >$ *with* K *attributes observed by* N *adversaries, the MAA leakage of* Δ *is bounded by*

$$\max_{1 \leq i \leq N} \mathcal{L}(\mathbf{A} \rightarrow Y_i) \leq \mathcal{L}(\mathbf{A} \rightarrow \mathbf{O}) \leq \frac{1}{K} \sum_{i=1}^{K} H_\infty(X_i). \tag{17}$$

The theorem states that the data leakage is at least as much as what is leaked to any single adversary, but no more than the average mean of all the attributes' min-entropy. In Example 5, the total leakage contains both the name and address, exceeding what any single adversary knows but remaining smaller than the sum of information leaked to each adversary separately. This is because Eve already knows the name, making the information leaked to Oscar irrelevant.

V. Capacity of MAA Leakage

The MAA leakage accounts for multiple attributes and adversaries in operational scenarios. However, its robustness depends on factors such as the prior π and the channel $P_{Y|X}$, which become more complex under MMA scenarios. To assess the reliability of this measure, we consider the concept of *capacity*, which represents the maximum leakage achievable across all possible priors π under a given set of channels.

Definition 7. *Given $\Delta =< \mathbf{A}, \mathbf{O} >$ with K attributes $\mathbf{A} = \{X_i \mid 1 \leq i \leq K\}$ and a prior distribution $\boldsymbol{\pi} = \{\pi_i \mid P_{X_i} = \pi_i,\ 1 \leq i \leq K\}$, observed by N adversaries $\mathbf{O} = \{Y_i \mid 1 \leq i \leq N\}$ through the channel $\mathbf{C} : \{\mathbf{C}_{i,j} = P_{X_i|Y_j} \mid 1 \leq i \leq K, 1 \leq j \leq N\}$, the capacity of $\mathbf{C}$ is defined as:*

$$\mathcal{ML}(\mathbf{A}, \mathbf{C}) = \sup_{\boldsymbol{\pi}} \log \frac{\mathcal{V}(\mathbf{A}|\mathbf{O})}{\mathcal{V}(\mathbf{A})}. \tag{18}$$

The capacity sets an upper limit on the potential leakage, independent of the prior distribution, based on the adversaries' channel set. By controlling the channel matrix $P_{Y|X}$, the system designer can offer a security guarantee.

Theorem 2. *The capacity of a single attribute data $\Delta =< X, \mathbf{O} >$, observed through N channels $\mathbf{C} : \{\mathbf{C}_i = P_{X|Y_i} \mid 1 \leq i \leq N\}$, is achieved with a uniform prior π:*

$$\mathcal{ML}(X, \mathbf{C}) = \log \sum_{\mathbf{O}} \max_x (\prod_{i=1}^{N} p(y_i|x)). \tag{19}$$

Theorem 3. *Given $\Delta =< \mathbf{A}, \mathbf{O} >$ with K attributes observed by N adversaries, the channel $\mathbf{C} : \{\mathbf{C}_{i,j} = P_{X_i|Y_j} \mid 1 \leq i \leq K, 1 \leq j \leq N\}$, and $\beta_i = 1/|X_i|$, the capacity of $\mathbf{C}$ is achieved when all attributes follow a uniform distribution. The maximum leakage occurs when there exists $\lambda \geq 0$ such that $\lambda = (\max_{x_i} p(\mathbf{o}|x_i))/(\max_{x_i} p(x_i))$ for all i.*

$$\mathcal{ML}(\mathbf{A}, \mathbf{C}) = \log \Big(|\mathcal{O}| \cdot \lambda \cdot (\sum_{i=1}^{K} \beta_i^2)/(\sum_{i=1}^{K} \beta_i) \Big). \tag{20}$$

Proof. From Cauchy-Schwarz inequality, for any list of $\{a_i, b_i \in \mathbb{R} | 1 \leq i \leq N\}$ we have:

$$(\sum_{i=1}^{K} a_i b_i)^2 \leq (\sum_{i=1}^{K} a_i^2)(\sum_{i=1}^{K} b_i^2).$$

$$\begin{aligned} \mathcal{L} &\leq \log \sum_{\mathbf{o}} \frac{\sum_{i=1}^{K} \max_{x_i} p(\mathbf{o}|x_i) \max_{x_i} p(x_i)}{\sum_{i=1}^{K} \max_{x_i} p(x_i)} \\ &\leq \log \frac{\sum_{\mathbf{o}} (\sum \max_{x_i} p(\mathbf{o}|x_i)^2 \sum \max_{x_i} p(x_i)^2)^{\frac{1}{2}}}{\sum \max_{x_i} p(x_i)} \\ &= \log \Big(|\mathcal{O}| \cdot \lambda \cdot (\sum_{i=1}^{K} \beta_i^2)/(\sum_{i=1}^{K} \beta_i) \Big). \end{aligned}$$ □

The numerical results of Theorem 3 are shown in Fig. 4. We simplify the computation by considering 10 attributes following a Bernoulli distribution with individual parameters q_i, and 10 adversaries observing them through individual binary symmetric channels (BSC) with error rates $0 \leq p_i \leq \frac{1}{2}$. The capacity is evaluated under uniform distributions with the same set of channels in each trial. The rate of correct guessing is assessed based on 100 observations made by 10 adversaries on 10 attributes in each of the 300 trials. The results demonstrate that the MAA leakage of the Bernoulli prior remains below its capacity with the same set of channels. The probability of correct guessing reflects the error rate in the BSCs for each trial.

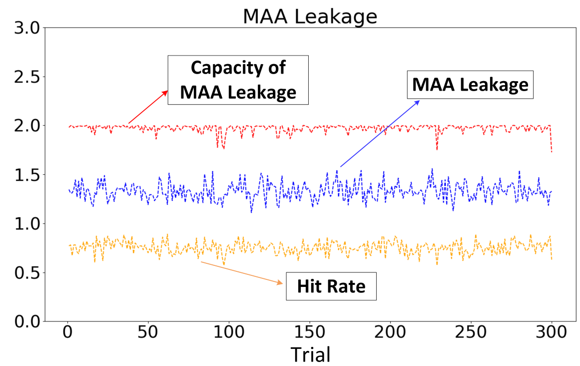

Fig. 4. Numerical results of Theorem 3.

VI. Conclusion

This paper pioneered a novel approach to analyze the requirements of leakage measures in the MAA scenario and introduces the MAA leakage. We validate its compliance with a leakage measure and operational interpretation. Additionally, the capacity of MAA leakage yields a closed-form expression, which provides a security guarantee for the system designer.

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